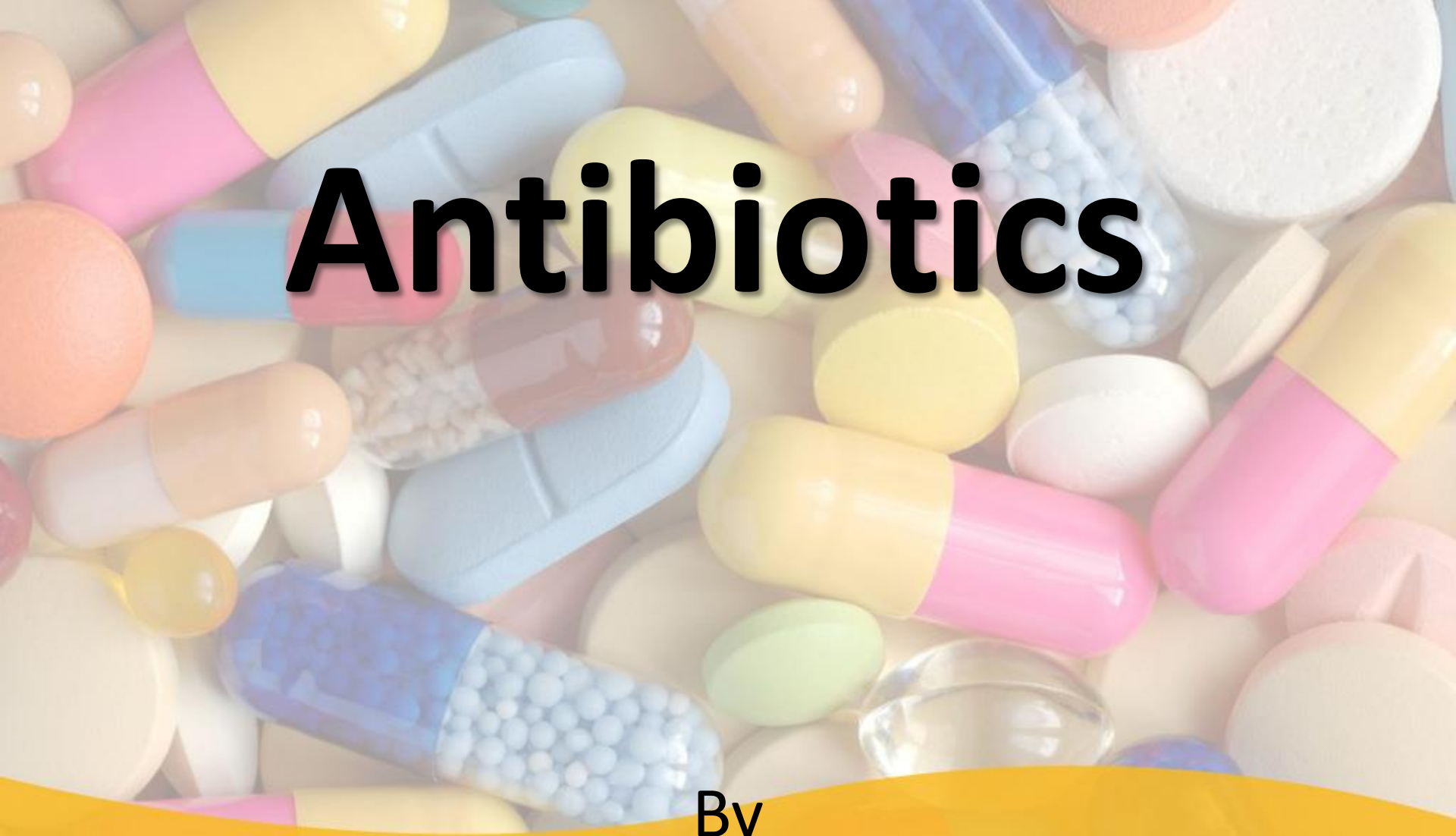


Antibiotics

By

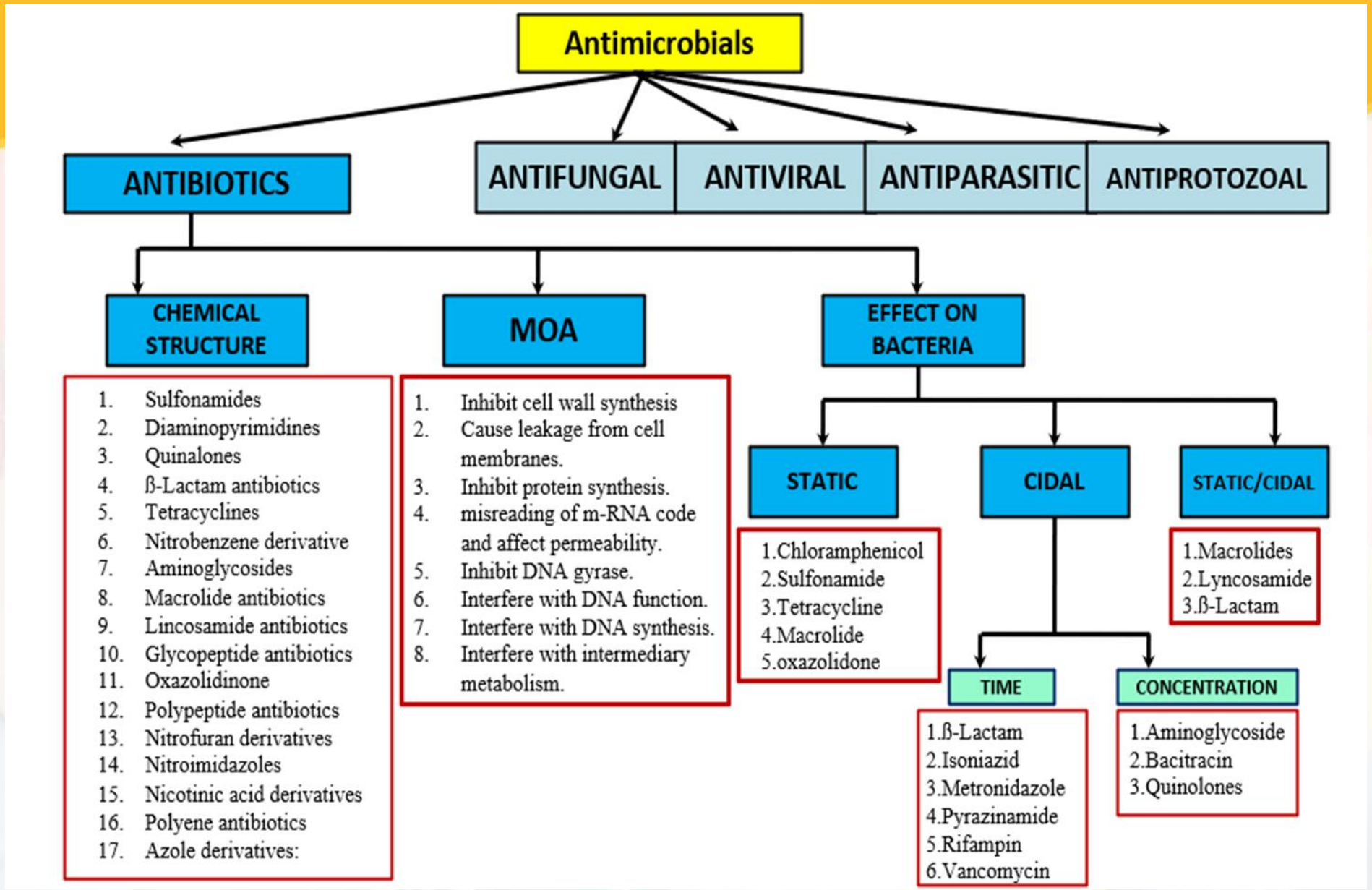
Ahmed Radwan

Lecturer of Tropical Medicine & Gastroenterology
Faculty of Medicine, Assiut University

What is an Antimicrobial (Antibiotic)?

"Antibiotic" is from antibiosis, meaning against life.

Substances derived from a microorganism or produced synthetically (Sulfonamides and Quinolones) to kill or suppress the growth of other microorganisms.


Antimicrobials
ANTIBIOTICS
ANTIFUNGAL
ANTIVIRAL
ANTIPARASITIC
ANTIPROTOZOAL
CHEMICAL STRUCTURE
MOA
EFFECT ON BACTERIA
1. Sulfonamides
2. Diaminopyrimidines
3. Quinalones
4. ß-Lactam antibiotics
5. Tetracyclines
6. Nitrobenzene derivative
7. Aminoglycosides
8. Macrolide antibiotics
9. Lincosamide antibiotics
10. Glycopeptide antibiotics
11. Oxazolidinone
12. Polypeptide antibiotics
13. Nitrofuran derivatives
14. Nitroimidazoles
15. Nicotinic acid derivatives
16. Polyene antibiotics
17. Azole derivatives:
1. Inhibit cell wall synthesis
2. Cause leakage from cell membranes.
3. Inhibit protein synthesis.
4. misreading of m-RNA code and affect permeability.
5. Inhibit DNA gyrase.
6. Interfere with DNA function.
7. Interfere with DNA synthesis.
8. Interfere with intermediary metabolism.
STATIC
CIDAL
STATIC/CIDAL
1.Chloramphenicol
2.Sulfonamide
3.Tetracycline
4.Macrolide
5.oxazolidone
1.Macrolides
2.Lyncosamide
3.ß-Lactam
TIME
CONCENTRATION
1.ß-Lactam
2.Isoniazid
3.Metronidazole
4.Pyrazinamide
5.Rifampin
6.Vancomycin
1.Aminoglycoside
2.Bacitracin
3.Quinolones

Classification of Antibiotics

Antibiotics are classified by several ways:

- On the basis of mode of action
- On the basis of spectrum of activity
- On the basis of mechanism of action
-

On the basis of mode of action

Bactericidal antibiotics:

- Kill bacteria
- e.g. Cephalosporin, Penicillin, Erythromycin, Aminoglycosides & Cotrimoxazole

Bacteriostatic antibiotics:

- Inhibit growth of susceptible bacteria, rather than killing them immediately; will eventually lead to bacterial death
- e.g. Tetracycline, Chloramphenicol, Erythromycin & Lincomycin

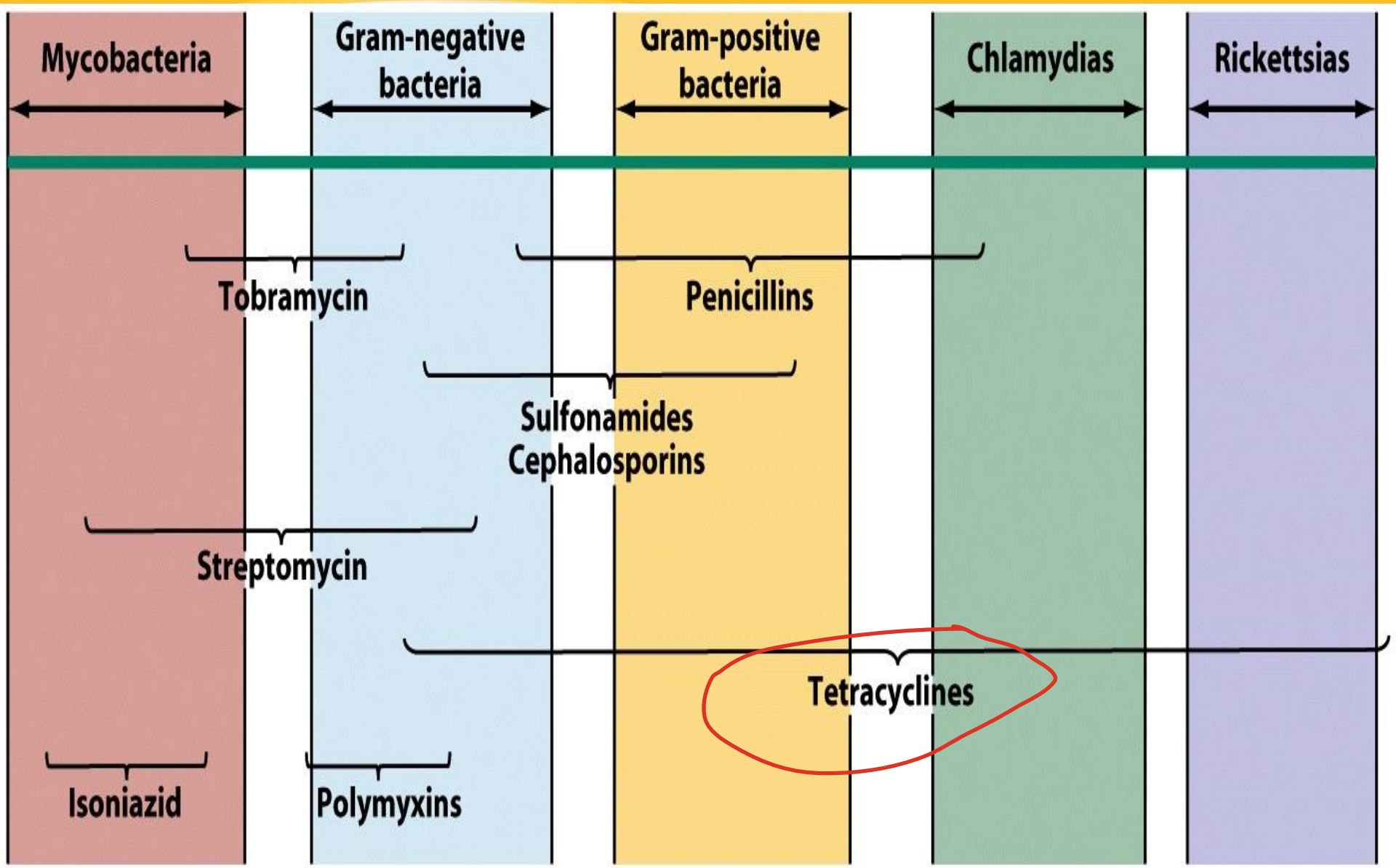
Antimicrobial Spectrum

Antimicrobial spectrum: the scope that a drug kills or suppresses the growth of microorganisms.

Narrow-spectrum: The drugs that only act on Small range (one strain of bacteria), e.g Isoniazide, Macrolids, Lincosamide, Glycopeptides)

Broad-spectrum: The drugs that have a wide antimicrobial scope.(Pencillins, Carbapenems, Tetracycline, Chloramphenicol)

On the basis of spectrum of activity



Mechanism of action of antimicrobial agents

1. Inhibition of cell wall synthesis:

- Penicillins, Cephalosporins, Bacitracin & Vancomycin

2. Inhibition of functions of cellular membrane:

- Polymyxins

3. Inhibition of protein synthesis:

- Chloramphenicol, Macrolides & Clindamycin
- Tetracyclines & Aminoglycosides

Mechanism of action of antimicrobial agents

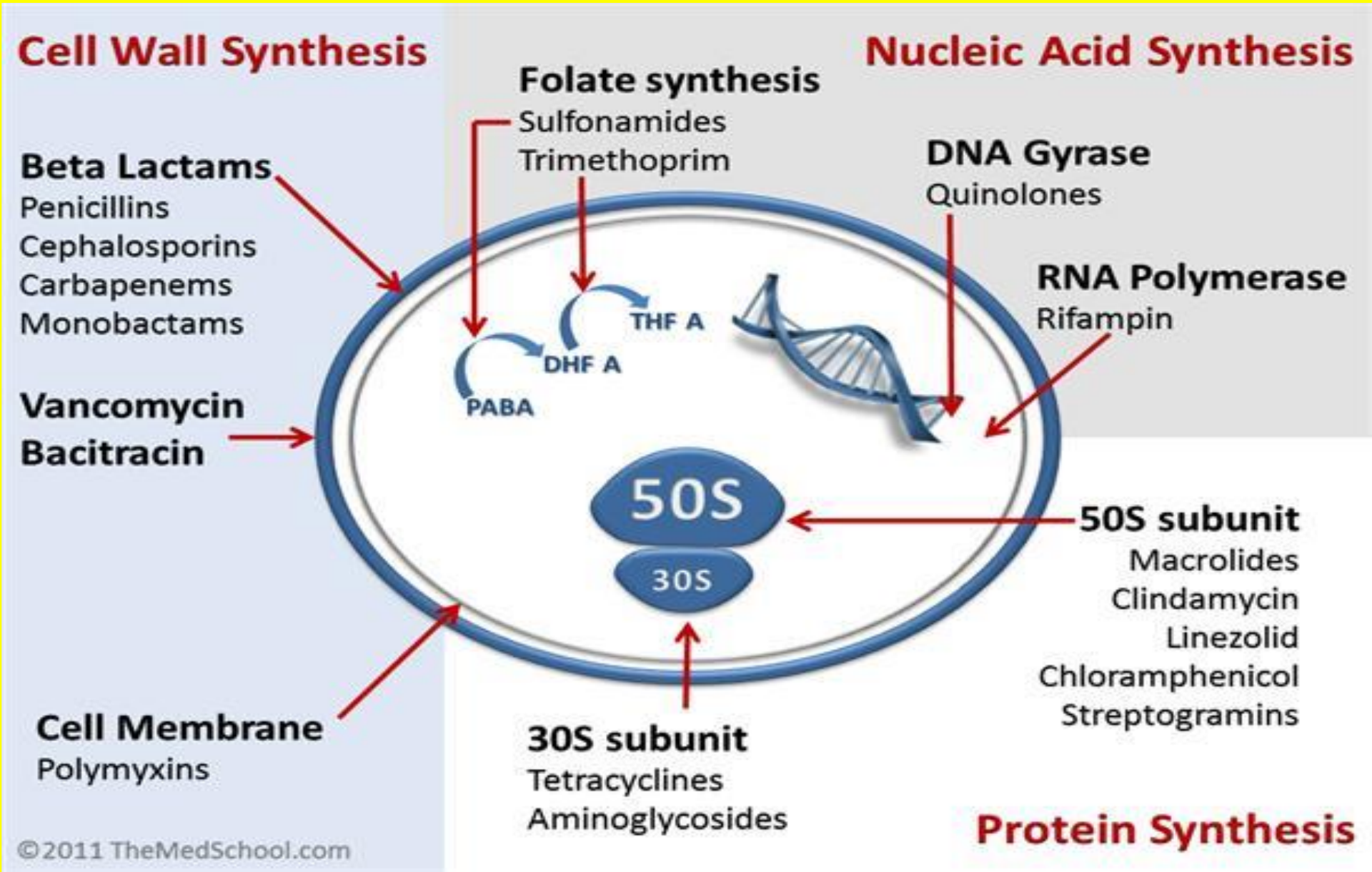
4. Inhibition of nucleic acid synthesis:

- Inhibition of RNA synthesis: Quinolones
- Inhibition of RNA synthesis: Rifampin

5. Inhibition of folic acid synthesis:

- Sulfonamides & trimethoprim

On the basis of mechanism of action



Misuse of Antibiotics

Antibiotic misuse, sometimes called antibiotic abuse or antibiotic overuse

The misuse or overuse of antibiotics, may produce serious effects on health.

It is a contributing factor to the creation of multidrug-resistant bacteria, informally called “super bugs”.

Antibiotic Resistances and Cross Resistances

Antibiotic resistance is the phenomenon that susceptibility of pathogenic microorganisms to antibiotic becomes lower or even loses after the microorganisms contact with antibiotic many times.

When the bacteria show resistance to one antibiotic, they are also resistant to some other antibiotics. This phenomenon is called cross antibiotic resistance.

Mechanisms of Antibiotic Resistance

1. Alteration of the target site of the antibiotic

- One of the most problematic antibiotic resistances worldwide, methicillin resistance among Staphylococcus aureus.

2. Enzyme inactivation of the antibiotic

- β-lactam antibiotics (Penicillins & Cephalosporins) can be inactivated by β-lactamases.

Mechanisms of Antibiotic Resistance Cont.

3. Active transport of the antibiotic out of the bacterial cell

- Active transport of the antibiotic out of the bacterial cell (efflux pumps) as removal of some antibiotics e.g. Tetracyclines, Macrolides, & Quinolones

4. Decreased permeability of the bacterial cell wall to the antibiotic

- Alteration in the porin proteins that form channels in the cell membrane e.g. Resistance of Pseudomonas aeruginosa to a variety of Penicillins & Cephalosporins

Mechanisms of Antibiotic Resistance



Decreased Permeability

β lactams
Amioglycosides

Efflux Pump

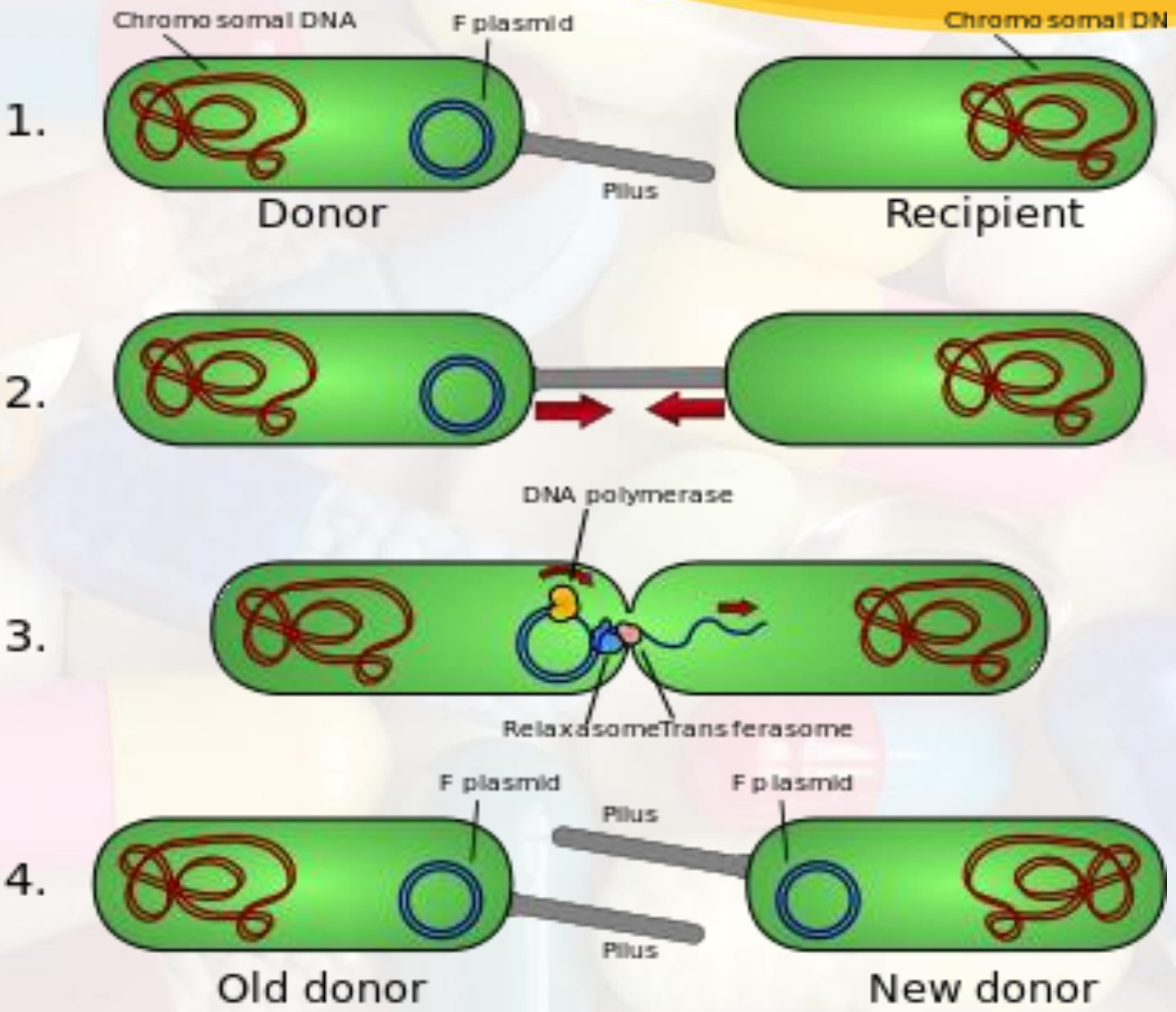
Tetracyclines
Quinolones
Macrolides

Altered Target

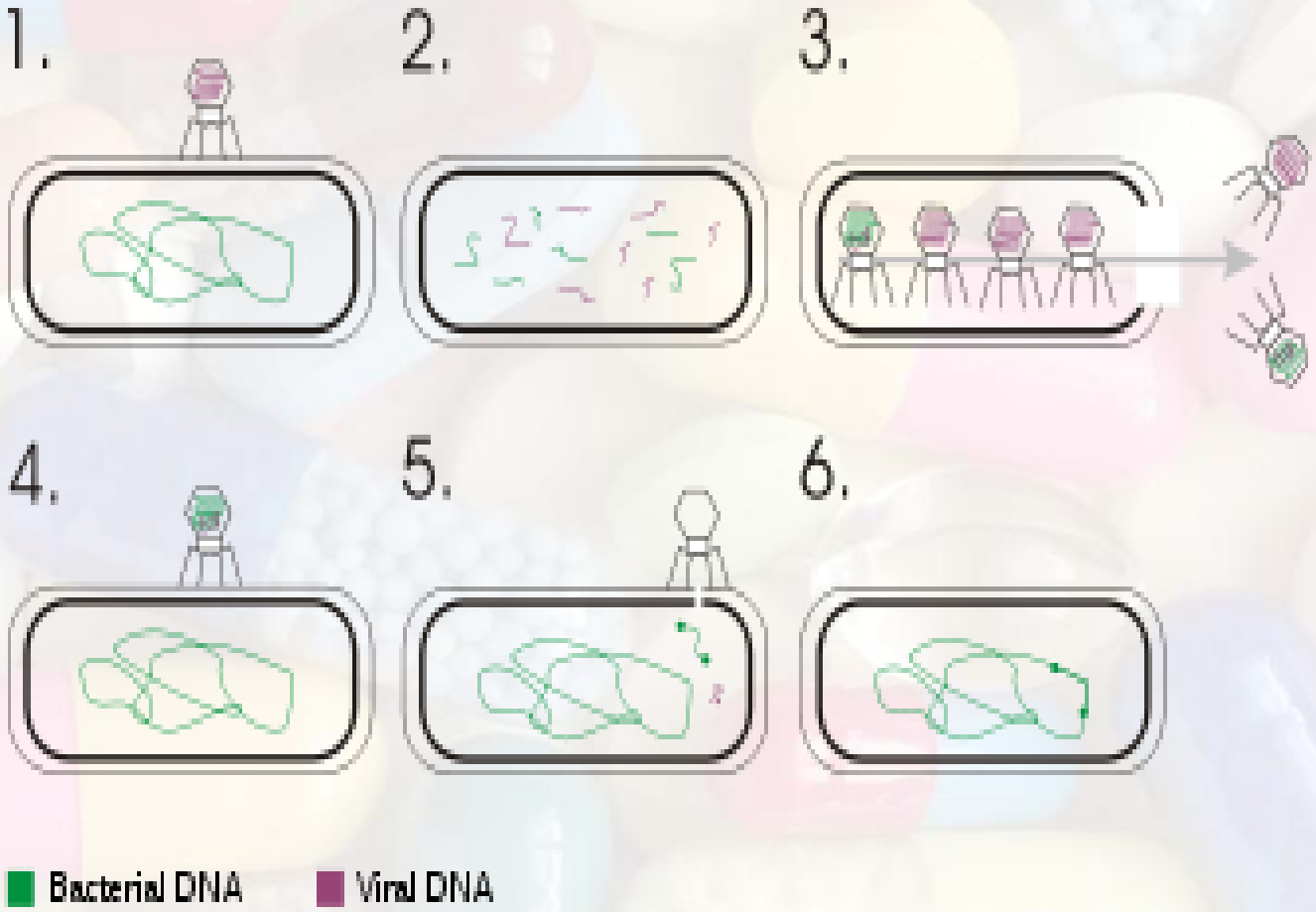
β lactams
Quinolones
Amioglycosides
Vancomycin
Macrolides
TMP/SMX

Enzymatic Inactivation

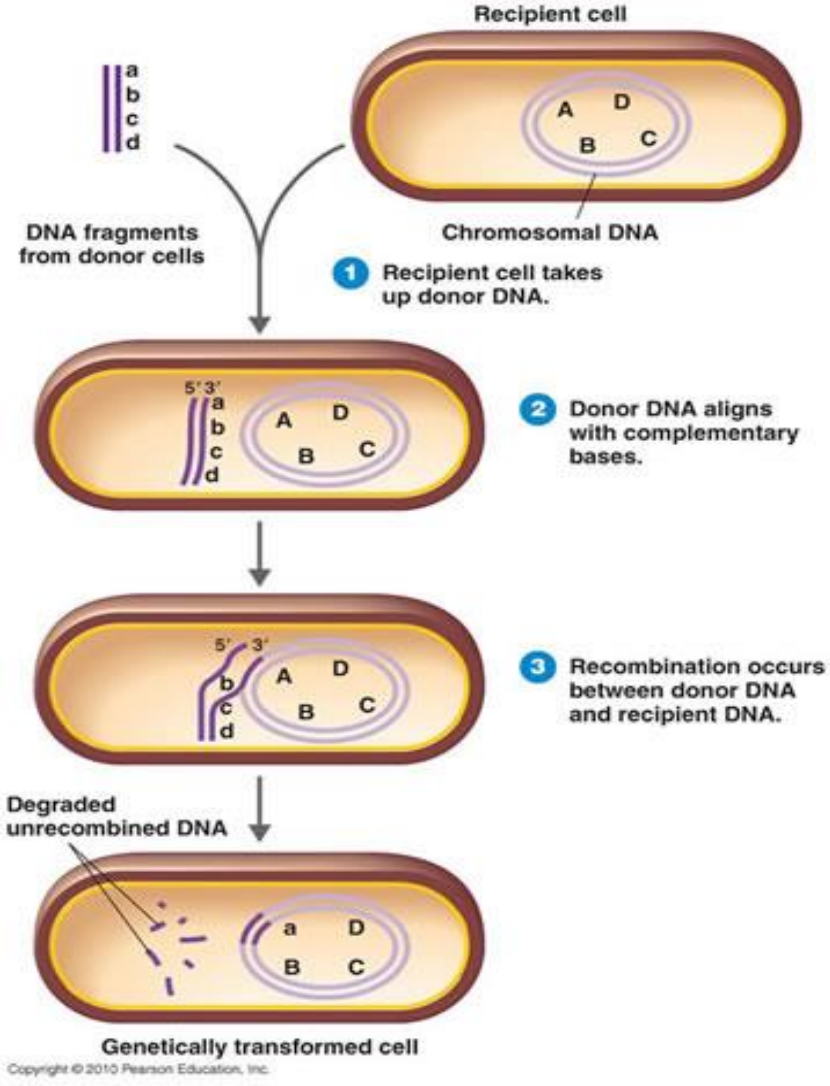
Conjugation



Transduction(bacterophage)



Transformation



General Principles of Antimicrobial Therapy

- Identification of the infecting organism should precede antimicrobial therapy when possible.

- The pathogenic microorganism susceptibility to antimicrobial agents should be determined, if a suitable test exists.

- Factors that influence the choice of an antimicrobial agent or its dosage for a patient include the age, renal and hepatic function, pregnancy status, and the site of infection, etc.

Selection of Antimicrobial Agent

- Empiric therapy - prior to identification of Organism - critically ill patients
- Organism's susceptibility to the antibiotic
- Patient factors - immune system, renal/hepatic function
- Effect of site of infection on therapy - blood brain barrier
- Safety of the agent
- Cost of therapy

Complications of Antibiotic Therapy

- Resistance due to inappropriate use of antibiotics
- Hypersensitivity (Penicillin)
- Direct toxicity (Aminoglycosides = ototoxicity)
- Super infections (broad spectrum antimicrobials cause alteration of the normal flora; often difficult to treat)

Antimicrobial Combination

Results of Combination Therapy:

- Additive (indifferent) effect: the activity of two drugs in combination is equal to the sum (or a partial sum) of their independent activity when studied separately.
- Synergistic effect: the activity of two drugs in combination is greater to the sum of their independent activity when studied separately.
- Antagonistic effect: the activity of two drugs in combination is less to the sum (or a partial sum) of their independent activity when studied separately.

Indications for the Clinical Use of Antimicrobial Combinations

- Prevention of the emergence of resistant organisms
- Polymicrobial infection
- Initial therapy
- Decreased toxicity
- Synergism

Prevention of the Emergence of Resistant Organisms

- Decreased resistant mycobacterium tuberculosis with combination
- Reduction of β-lactamase induction with combination β-lactam agents and Aminoglycosides

Polymicrobial Infection

- Intra-abdominal infection
- Pelvic infection
- Mixed aerobic and anaerobic organism

Initial Therapy

- Neutropenic patients
- In patients where the nature of infection is not clear yet

Decreased Toxicity

- Decrease the toxic drug required for treatment and thus reduce the dose related toxicity

Synergism

Enhanced Uptake of Aminoglycosides when Combined with β-lactam agents

- Treatment of enterococcal endocarditis: ampicillin & gentamicin
- Viridans streptococcal endocarditis: penicillin & gentamicin
- Staphylococcal bacteremia: vancomycin & gentamicin
- Treatment of pseudomonas infections: β-lactam agent & Aminoglycosides

Inhibition of Sequential Steps

- Sulfonamide with trimethoprim
- Treatment and prevention of chronic urinary tract infection, typhoid fever, shigellosis caused by organisms resistant to ampicillin

Inhibition of enzymatic activation

Disadvantages of the Inappropriate Use of Antimicrobial Combination

- Antagonism
- Increased cost
- Adverse effects
- Super infections (alteration of the normal flora; often difficult to treat)

Take a long breath…

Classes of Antibiotics

The main classes of antibiotics are:

- Beta-Lactams
  - Penicillins
  - Cephalosporins
  - Carbapenems
  - Monobactams
- Macrolides & Ketolides
- Aminoglycosides
- Fluoroquinolones
- Tetracyclines, Amphenicols

Penicillins

- The penicillins are the oldest class of antibiotics. Penicillins have a common chemical structure which they share with the Cephalosporins.
- Penicillins are generally bactericidal, inhibiting formation of the cell wall. Penicillins are used to treat skin infections, dental infections, ear infections, respiratory tract infections, urinary tract infections, gonorrhea.

Types of penicillins

- ***The natural penicillins***

  Benzylpenicillin (Penicillin G), Procaine Benzylpenicillin, Benzathine penicillin & Penicillin V

- ***Penicillinase-resistant penicillins***

  Methicillin & Oxacillin, flucloxacillin

- ***Aminopenicillins***

  Ampicillin & Amoxicillin

Types of penicillins

- ***Ureidopenicillins (long acting)***

  Piperacillin

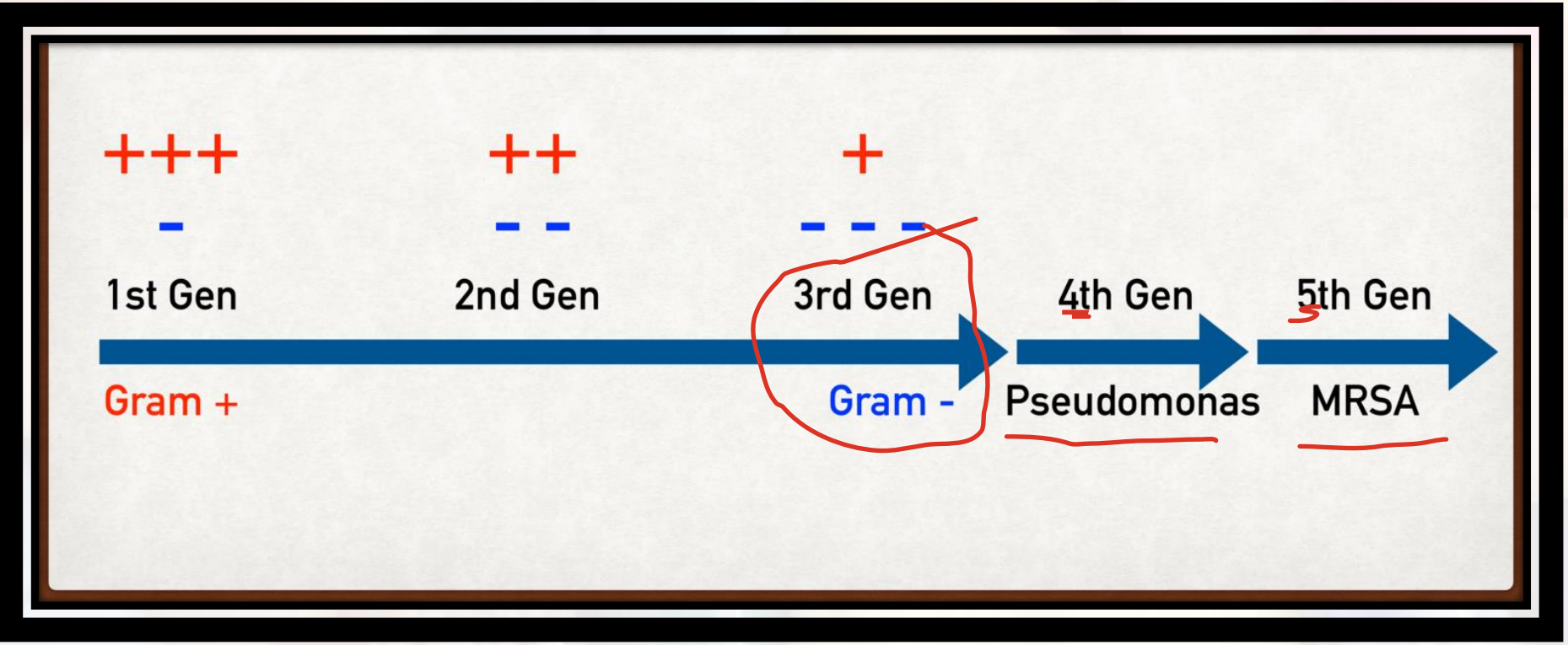
- ***Penicillins/Beta-lactamase inhibitors***

  Ampicillin/sulbactam, Amoxicillin/clavulanic, Piperacillin/tazobactam

Cephalosporins

1st generation	2nd generation	3rd generation	4th generation	5th generation
•Cephradine •Cephalexin •Cefadroxil •Cefazolin	•Cefoxitin •Cefotetan Cefaclor Cefuroxime	•Ceftazidime •Cefixime •Cefoperazone •Cefotaxime •Ceftriaxone •Cefdinir •Ceftibuten	•Cefepime •Cefpirome	•Ceftaroline •Ceftobiprole

Spectrum of Cephalosporins



Major Differences between Cephalosporin Generations

Generation	Activity against Gram Negative	Resistance to Beta-Lactamases	Distribution to CSF
1st (e.g., cephalothin)	Low	Low	Poor
2nd (e.g., cefamandole)	Higher	Higher	Poor
3rd (e.g. cefotaxime)	Higher	Higher	Good
4th (e.g. cefepime)	Highest	Highest	Good

Lehne, 2007, *Pharmacology for Nursing Care*, 6th ed., Elsevier, p. 973

Carbapenems

- a class of β-Lactam antibiotics with a broad spectrum of antibacterial activity
- Highly resistant to most β-lactamases
- Active against both Gram-positive and Gram-negative bacteria, and anaerobes, with the exception of intracellular bacteria (atypical), such as the *Chlamydia*.

- Agents:

  Imipenem/Cilastatin (Tienem)

  Meropenem (Meronem)

  Ertapenem

Monobactams

- They are β-lactam compounds
- They work only against aerobic Gram negative bacteria (Neisseria, Pseudomonas).
- The only commercially available Monobactams antibiotic is Aztreonam (Azactam).
- Adverse effects to Monobactams can include skin rash and occasional abnormal liver functions.

Glycopeptides

- **Glycopeptides** (e.g. vancomycin, teicoplanin)
- These antibiotics are effective principally against Gram-positive cocci.
- This class of drugs inhibit the synthesis of cell walls in susceptible microbes by inhibiting peptidoglycan synthesis.
- Vancomycin is used if infection with methicillin-resistant *Staphylococcus aureus* (MRSA) is suspected.

- Due to their toxicity, use of glycopeptides antibiotics is restricted to patients who are critically ill, who have a demonstrated hypersensitivity to the β-lactams, or who are infected with β-lactam-resistant species.
- They exhibit a narrow spectrum of action and are bactericidal only against the enterococci.
- Some tissues are not penetrated very well by glycopeptides and they do not penetrate into the cerebrospinal fluid.

- Oral preparations of vancomycin are available, however they are not absorbed from the lumen of the gut, formulated for the treatment of infections within the gastrointestinal tract, Clostridium difficile, for example.

Macrolides

- The Macrolides are bacteriostatic, binding with bacterial ribosomes to inhibit protein synthesis.
- Macrolides antibiotics are used to treat respiratory tract infections (such as pharyngitis, sinusitis, and bronchitis), genital, gastrointestinal tract, and skin infections.

- Macrolides antibiotics are:

  Erythromycin (Erythrocin)

  Clarithromycin (Klacid)

  Azithromycin (Zithromax)

  Roxithromycin (Roxicin)

Ketolides

- Antibiotics belonging to the Macrolides group
- Much broader spectrum than other Macrolides.
- Ketolides are effective against Macrolides-resistant bacteria, due to their ability to bind at two sites at the bacterial ribosome as well as having a structural modification that makes them poor substrates for efflux-pump mediated resistance.
- The only Ketolide on the market at this moment is telithromycin, which is sold under the brand name of Ketek.

Lincosamides

- Lincosamides (e.g. lincomycin, clindamycin) are a class of antibiotics.
- Lincosamides prevent bacteria replicating by interfering with the synthesis of proteins.

- They are normally used to treat staphylococci and streptococci, and have proved useful in treating Bacteroides fragilis and some other anaerobes.
- They are used in the treatment of Toxic Shock Syndrome.
- Lincosamides antibiotics are one of the classes of antibiotics most associated with pseudomembranous colitis caused by C. difficile.

Aminoglycosides

- Aminoglycosides antibiotics are used to treat infections caused by Gram-negative bacteria.
- Aminoglycosides may be used along with penicillins or cephalosporins to give a two-pronged attack on the bacteria.
- Since Aminoglycosides are broken down easily in the stomach, they can't be given by mouth and must be injected.

- Generally, Aminoglycosides are given for short time periods.

- Aminoglycosides group includes:

  Amikacin (Amikin)

  Gentamicin (Garamycin)

  Neomycin (Neomycin)

  Streptomycin (Streptomycin)

  Tobramycin (Nebcin)

Fluoroquinolones

- They are synthetic antibiotics and not derived from bacteria.
- Because of their excellent absorption Fluoroquinolones can be administered not only by intravenous but orally as well.
- Fluoroquinolones are used to treat most common urinary tract infections, skin infections, and respiratory infections (such as sinusitis, pneumonia, bronchitis).

QUINOLONE GENERATIONS (PHARMACODYNAMIC CLASSIFICATION)

First	Second	Third	Fourth
Nalidixic Acid	Lomefloxacin	Levofloxacin	Trovafloxacin
	Norfloxacin	Sparifloxacin	Moxifloxacin
	Enoxacin	Grepafloxacin	Gatifloxacin
	Ofloxacin		Clinafloxacin
	Ciprofloxacin		Sitafloxacin
	Fleroxacin		Gemifloxacin

Trimethoprim/sulfamethoxazole or Co-trimoxazole

- An antibiotic used in the treatment of a variety of bacterial, fungal and protozoal infections.
- Co-trimoxazole is generally considered bactericidal, although its components are individually bacteriostatic.
- Its actions are antifolate in nature, inhibiting both *de novo* folate biosynthesis and metabolism.

Tetracyclines

- Broad-spectrum bacteriostatic agents and work by inhibiting the bacterial protein synthesis.
- Tetracyclines may be effective against a wide variety of microorganisms, including Rickettsia and Amoebic parasites.
- Tetracyclines are used in the treatment of infections of the respiratory tract, sinuses, middle ear, urinary tract, skin, intestines.

- Tetracyclines also are used to treat Gonorrhea, Rocky Mountain spotted fever, Lyme Disease, Typhus.
- Their most common current use is in the treatment of moderately severe acne and rosacea.
- Tetracycline antibiotics are:

  Tetracycline

  Doxycycline (Vibramycin)

  Minocycline

  Oxytetracycline (Oxytetracid)

  Tigacycline

Chloramphenicol

- It is a bacteriostatic
- It is considered a prototypical broad-spectrum antibiotic, alongside the tetracyclines, and as it is both cheap and easy to manufacture, it is frequently an antibiotic of choice in the developing world.
- Effective against a wide variety of Gram-positive and Gram-negative bacteria, including most anaerobic organisms.

- Due to resistance and safety concerns, it is no longer a first-line agent for any infection in developed nations, with the notable exception of topical treatment of bacterial conjunctivitis.

Oxazolidinones (e.g. linezolid)

- Linezolid inhibits ribosomal protein synthesis by inhibiting formation of the initiation complex.
- It is used for the treatment of serious infections caused by Gram-positive bacteria that are resistant to several other antibiotics.
- Linezolid is active against most Gram-positive bacteria that cause disease, including streptococci, vancomycin-resistant enterococci (VRE), and methicillin-resistant *Staphylococcus aureus* (MRSA).

Ansamycins

•A family of secondary metabolites that show antimicrobial activity against many Gram-positive and some Gram-negative bacteria and includes various compounds, e.g. streptovaricins and rifamycins

• among Rifamycins are a subclass of ansamycins with high potency against mycobacteria. This resulted in their widespread use in the treatment of tuberculosis, leprosy, and AIDS-related mycobacterial infections.

- The antibacterial activity of rifamycins relies on the inhibition of bacterial DNA-dependent RNA synthesis. This is due to the high affinity of rifamycins for the prokaryotic RNA polymerase.

Rifaximin is a semisynthetic antibiotic basedon rifamycin. It has poor oral bioavailability, meaning that very little of the drug will be absorbed into the blood stream when it is taken orally. Rifaximin is used in the treatment of traveler's diarrhea and hepatic encephalopathy.

Rifaximin interferes with transcription by binding to the β-subunit of bacterial RNA polymerase.

Metronidazole

- A nitroimidazole antibiotic medication used particularly for anaerobic bacteria and protozoa.

- It is antibacterial against anaerobic organisms, an amoebicide, and an antiprotozoal.

- It inhibits nucleic acid synthesis by disrupting the DNA of microbial cells. This function only occurs when Metronidazole is partially reduced, and because this reduction usually happens only in anaerobic cells, it has relatively little effect upon human cells or aerobic bacteria.

- Metronidazole is primarily used to treat: bacterial vaginosis, pelvic inflammatory disease (along with other antibacterials like ceftriaxone), pseudomembranous colitis, aspiration pneumonia, rosacea (topical), fungating wounds (topical), intra-abdominal infections, lung abscess, gingivitis, amoebiasis, giardiasis, trichomoniasis, and infections caused by susceptible anaerobic organisms such as *Bacteroides.*

- It is also often used to eradicate *Helicobacter pylori* along with other drugs and to prevent infection in people recovering from surgery.

- Adverse effects include:

  Discolored urine, headache, metallic taste , nausea

  Metronidazole is classed as pregnancy category B

Polymyxins

- The most popular : Colisitin
- Mechanism of Action: Binds to lipopolysaccharides (LPS) and phospholipids in the outer membrane of Gram-negative bacteria.Disrupts membrane integrity, leading to cell lysis and bacterial death.
- Active against Gram-negative bacteria only Highly effective against multidrug-resistant (MDR) pathogens, including:Acinetobacter baumanniiPseudomonas aeruginosaKlebsiella

- treatment of MDR Gram-negative infections, including:Ventilator-associated pneumonia (VAP)BacteremiaComplicated urinary tract infections (cUTIs)Intra-abdominal infections

Antibiotic Class	Antibiotic Class Members	Most Common Side Effects
Penicillins	penicillin, amoxicillin, amoxicillin-clavulanate, ampicillin, piperacillin-tazobactam, nafcillin, oxacillin	**Rash, diarrhea**, abdominal pain, nausea/vomiting, drug fever, **hypersensitivity (allergic) reactions**
Cephalosporins	cephalexin, cefaclor, cefuroxime, ceftibuten, cefdinir, cefixime, ceftriaxone	**Rash, diarrhea,** nausea/vomiting (rare), **hypersensitivity (allergic) reactions**, serum sickness,
Carbapenems	meropenem, ertapenem, doripenem, imipenem-cilastatin	Diarrhea, nausea/vomiting, headache, rash, **liver toxicity, eosinophilia** (elevated white blood cells)

Antibiotic Class	Antibiotic Class Members	Most Common Side Effects
Aminoglycosides	gentamicin, tobramycin, amikacin	**Nephrotoxicity, ototoxicity** (hearing loss), dizziness, nausea/vomiting, nystagmus
Macrolides	erythromycin, azithromycin, clarithromycin	Abdominal pain, diarrhea, anorexia, nausea/vomiting, **hepatotoxicity** (erythromycin)
Tetracyclines	tetracycline, doxycycline, minocycline	Nausea/vomiting, diarrhea, sore mouth, abdominal pain, **tooth discoloration in children < 8 years, liver toxicity**

Antibiotic class	Antibiotic Class Members	Most Common Side Effects
Sulphonamides	trimethoprin-sulfamethoxazole, sulfadiazine	nausea/vomiting, diarrhea, anorexia, abdominal pain, rash, **photosensitivity,** headache, **hemolytic anemia**
Lincosamide	clindamycin, lincomycin	**Pseudomembranous colitis** (may be severe), diarrhea, nausea/vomiting, rash, hypersensitivity, jaundice (clindamycin)
Quinolones	Ciprofloxacin, norfloxacin,levofloxacin	Nausea/vomiting, headache, **arthropathy in children,nephrotoxicity**

Antibiotic class	Antibiotic Class Members	Most Common Side Effects
Glycopeptides	Vancomycin	**Red man syndrome (flushing, hypotension, itching); phlebitis, nephrotoxicity** nausea/vomiting, headache, dizziness
Miscellanous	Metronidazole	Nausea/vomiting, dizziness, headache, vaginal candidiasis, **metallic taste, peripheral neuropathy, discolored urine**
Chloramphenicol	chloramphenicol	**Grey baby syndrome, bone marrow aplasia**

Antibiotic class	Antibiotic Class Members	Most Common Side Effects
Oxazolidinones	Linezolid	bone marrow suppression and low platelet counts, peripheral neuropathy and optic nerve damage, and lactic acidosis.
Ansamycins	Rifamycins	Nausea/vomiting, abdominal pain, **Hepatotoxicity**, flushing, pruritus, rash, redness and watering of eyes
Polymyxcin	Colistin	**Nephrotoxicity and neurotxicity**

Category B	Category C	Category D
Penicillin	Clarithromycin	Aminoglycoside
Cephalosporin	Quinolones	Tetracyclines
Azithromycin, Erythromycin.	Carbapenems	Sulfonamides
Nitrofurantoin (avoid at term)	Vancomycin	Chloramphenicol
Metronidazole		

Streptogramins

- Effective in the treatment of vancomycin-resistant *Staphylococcus aureus* (VRSA) and vancomycin-resistant *Enterococcus* (VRE), two of the most rapidly growing strains of multidrug-resistant bacteria.
- Members include:

  Quinupristin/dalfopristin

  Pristinamycin

  Virginiamycin

Adverse effects include:

Related to administration via peripheral vein

Inflammation, pain, edema, infusion site reaction, thrombophlebitis

Non-venous adverse effects

- Nausea, vomiting, diarrhea
- Rash
- Headache
- Pain, ill-defined focal or generalized discomfort
- Pruritus
- Arthralgia, myalgia
- Asthenia
- Conjugated hyperbilirubinaemia

New drugs

Plazomicin

Eravacycline

Cefiderocol

Antibiotics that combine **BL-BLIs**
Ceftazidime /avibactam
Ceftolozane /tazobactam
Meropenem /vaborbactam,
Imipenem-cilastatin / relebactam.

Temocillin only approved in Belgium and the United Kingdom

Plazomicin

- Brand name (**Zemdri)**
- **Mechanism of action** : A synthetic aminoglycoside, Inhibits bacterial protein synthesis.
- **Used in** complicated urinary tract infections (cUTIs) and pyelonephritis (FDA approved in 2018) .
- **Side effects** : possibility of nephrotoxicity and ototoxicity
- **Activity** : plazomicin is active against > 95% of Enterobacterales isolates , against ESBL isolates, and against carbapenem-resistant isolates

Eravacycline

- Brand name (**Xerava**)
- **Mechanism of action** A fluorocycline of the tetracycline class ,it inhibits bacterial protein synthesis.
- **Used in** treatment of cIAI at for a total duration of 4 to 14 days . (FDA approved in 2018)
- In **Egypt (Nuzyra = Omadacyclin)**
  - **Side effects** : as other tetracyclines, permanent tooth discoloration may occur .
  - **Activity** : against E. coli , K. pneumoniae - Acinetobacter spp and CRE isolates .

Temocillin

- Brand name **Negaban ; Temopen**
- **Mechanism of action** Temocillin is a semisynthetic derivative of ticarcillin , a penicillin antibiotic targeting penicillin-binding protein (PBP3) , inhibiting the synthesis of the peptidoglycan layer of bacterial cell wall .
- **Used in** alternative to carbapenems treating infections due to ESBL-producing *Enterobacterales* and uncomplicated UTI due to Klebsiella pneumoniae Carbapenemase producers .
- **Side effects** as injectable penicillins , pain at the site of I.M. injection, occasionally urticarial rash .
- **Activity** against ESBL, and carbamenase producing Enterobacterales .

Cefiderocol

- Brand name (**Fetroja)**
- **Mechanism of action** : a novel combination of a catechol-type siderophore and a cephalosporin antibiotic , binding to and inhibiting penicillin-binding proteins (PBPs) ,this structure enhanced stability against hydrolysis by many β-lactamases
- **Used in** treatment of cUTI, (FDA approved in 2019) and this indication was extended to HAP/VAP in 2020.
- **Side effects** infusion site reactions, candidiasis, cough, elevations in liver tests, hypokalaemia, hypomagnesemia, and atrial fibrillation.

(Cefiderocol has FDA label warning for higher mortality versus other antibiotics in critically ill patients with multidrug-resistant Gram-negative bacterial infections)

- **Activity** against of Enterobacterales isolates (ESBL) ,

Beta-Lactam/Beta-Lactamase Inhibitor

These antibiotics are a combination of a cephalosporin or a carbapenem antibiotic with a beta-lactamase inhibitor (BLI).

Mechanism of action inhibiting beta-lactamases, the partner beta-lactam antibiotic is allowed to reach its target, the penicillin-binding proteins (PBPs).

Ceftazidime/Avibactam

- Brand name (**Avycaz)**
- **Used in** treating complicated intra abdominal infections (cIAI) (in combination with metronidazole) and cUTI (FDA approvedin 2015) extended to HAP/VAP in 2018**.**
- **Side effects** positive direct Coombs test.
- **Activity** against Gram-negative bacteria producing ESBL, carbapenem non susceptible and P. aeruginosa .
  - **In Egypt (Zavicefta)**
  - **Used in** complicated intra-abdominal infections (cIAI); Complicated urinary tract infections (cUTI), including pyelonephritis; hospital-acquired pneumonia (HAP), including ventilator associated pneumonia (VAP)

Ceftolozane/Tazobactam

- Brand name (**Zerbaxa)**
- It is an anti-pseudomonal cephalosporin partnered with BLI tazobactam.
- **Used in** : cUTI and cIAI.(FDA approved in 2014) .The indication was extended to HAP/VAP in 2019.
- **Activity** against ESBL Enterobacterales and carbapenem non susceptible P. aeruginosa .

Meropenem/Vaborbactam

- Brand name (**Vabomere)**
- **Used in** FDA approved to treat cUTI at a dose of 4 g (meropenem 2 g and vaborbactam 2 g) .
- **Side effects** occurring in 3% of patients were phlebitis /infusion-site reactions .
- **Activity** against carbapenem resistence Enterobacterales (CRE)

Imipenem-Cilastatin/Relebactam

- Brand name (**Recarbrio)**
- **Used in** cUTI and cIAI (FDA approved in 2019) (imipenem 500 mg, cilastatin 500 mg, and relebactam 250 mg). The indication was extended to HAP/VAP in 2020.
- **Side effects** elevated lever enzymes, increased eosinophils, rash.
- **Activity** against Enterobacterales (including ESBL) and P. aeruginosa , K. pneumoniae isolates producing carbamenases.
- Because imipenem is rapidly inactivated by renal dehydropeptidase I (DHP-1), it is given in combination with cilastatin, a DHP-I inhibitor which increases half-life and tissue penetration of imipenem

For more information:

- http://www.emedexpert.com/classes/antibiotics.shtml
- http://en.wikipedia.org/wiki/List_of_antibiotics
- http://www.orthobullets.com/basic-science/9059/antibiotic-classification-and-mechanism
- http://www.nlm.nih.gov/medlineplus/antibiotics.html
- https://www.youtube.com/channel/UCvl8mk9dKfhidf8oVhYkaOg

Thanks

